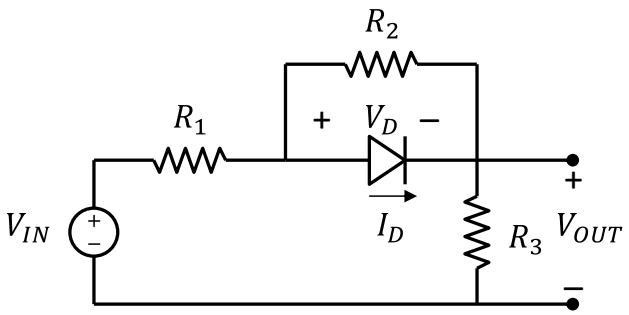
3 Diodi

Esercizio 3.1

DATI: $R_1 = 1k\Omega$, $R_2 = 2k\Omega$, $R_3 = 1k\Omega$, $V_{ON} = 0$ (diodo ideale).

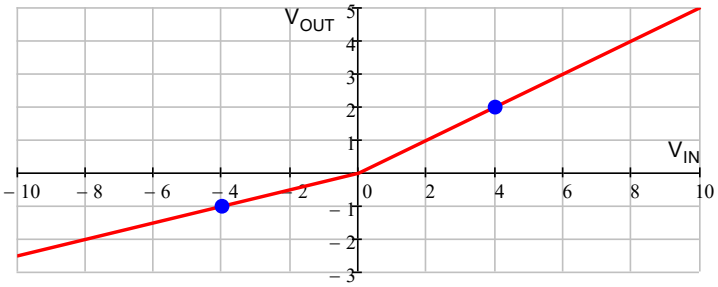
Dato il circuito in figura:

1. Calcolare la tensione di uscita e la polarizzazione del diodo con $V_{IN} = 4V$
2. Calcolare la tensione di uscita e la polarizzazione del diodo con $V_{IN} = -4V$
3. Tracciare la transcaratteristica $V_{OUT} = f(V_{IN})$.



Soluzione

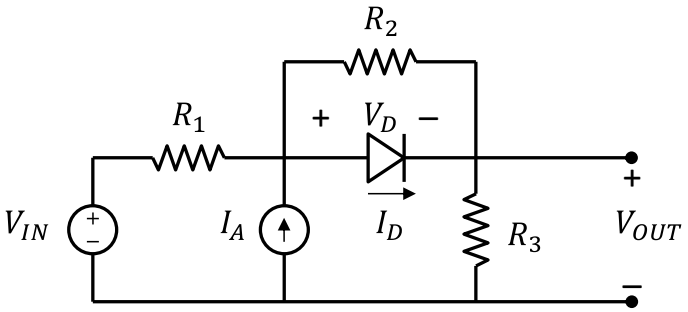
1. $V_{OUT} = 2V$ (diodo ON, $I_D = 2mA$)
2. $V_{OUT} = -1V$ (diodo OFF, $V_D = -2V$)
3. Transcaratteristica



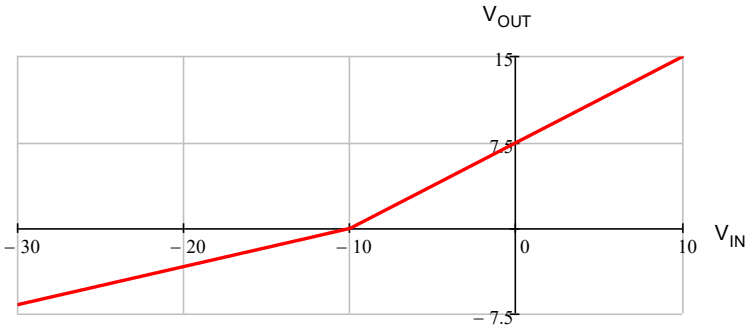
Esercizio 3.2

DATI: $R_1 = 1k\Omega$, $R_2 = 2k\Omega$, $R_3 = 3k\Omega$, $I_A = 10mA$, $V_{ON} = 0$ (diodo ideale).

Dato il circuito in figura tracciare la transcaratteristica $V_{OUT} = f(V_{IN})$.



Soluzione

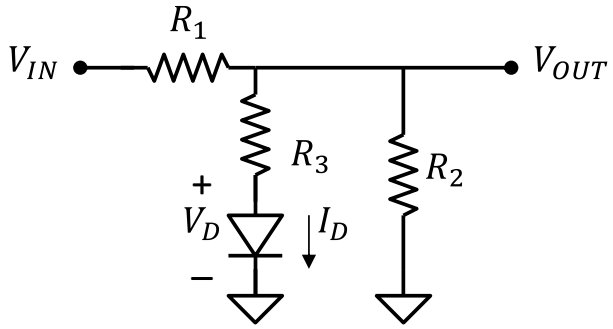


Esercizio 3.3

DATI: $R_1 = 8k\Omega$, $R_2 = 8k\Omega$, $R_3 = 1k\Omega$, $V_{ON} = 0.5V$

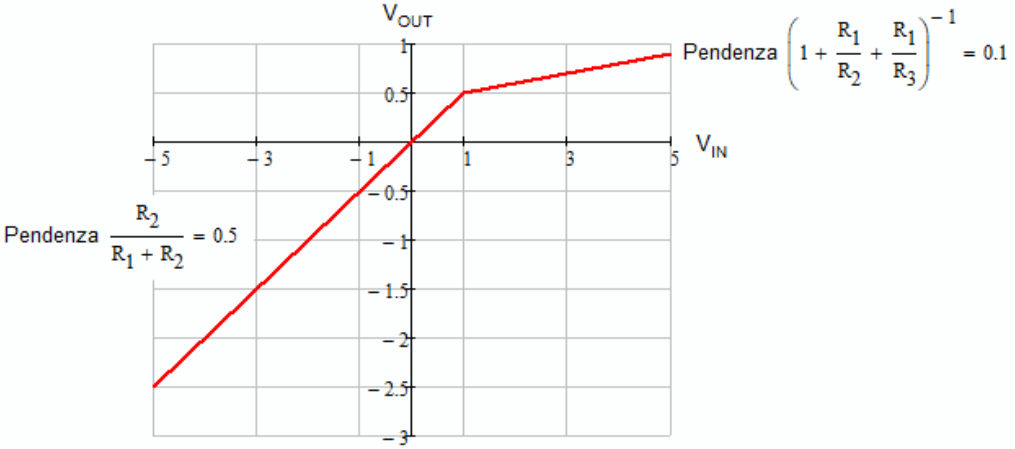
Dato il circuito in figura formato da tre resistenze e un diodo

1. Tracciare la transcaratteristica della tensione V_{OUT} in funzione della tensione V_{IN}.
2. Calcolare la tensione V_{OUT} corrispondente al punto in cui il diodo cambia regione operativa



Soluzione

1. Transcaratteristica

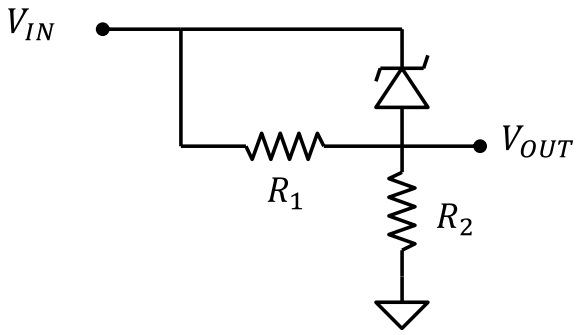


2. $V_{IN} = 1V$, $V_{OUT}=0.5V$

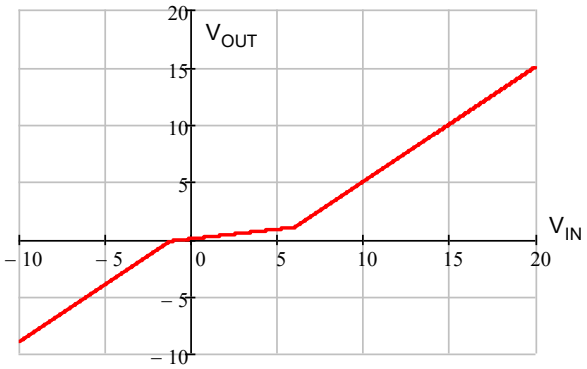
Esercizio 3.4

DATI: $R_1 = 5k\Omega$ $R_2 = 1k\Omega$, $V_{ON} = 1V$, $V_Z = 5V$.

Tracciare la transcaratteristica del circuito in figura.



Soluzione

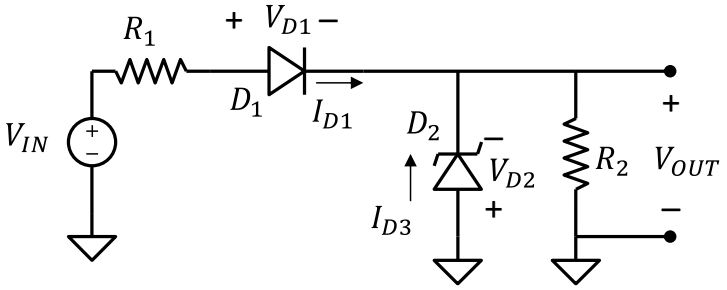


Esercizio 3.5

DATI: R_1 = 1kΩ R_2 = 3.6kΩ, V_{ON} = 0.7V, V_Z = 5.4V

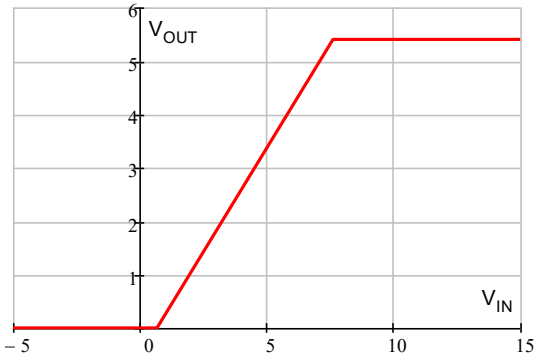
Dato il circuito in figura:

1. Calcolare la tensione V_{OUT} con V_{IN} = 10V
2. Calcolare la tensione V_{OUT} con V_{IN} = -10V
3. Calcolare la tensione V_{OUT} con V_{IN} = 3V
4. Tracciare la transcaratteristica.



Soluzione

1. V_{OUT} = 5.4V
2. V_{OUT} = 0V
3. V_{OUT} = 1.8V
4. Transcaratteristica

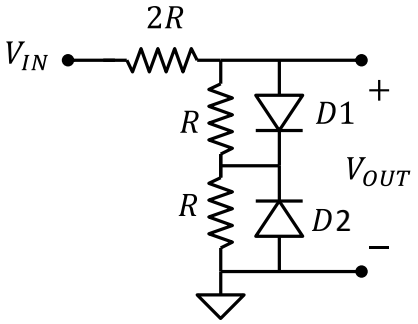


Esercizio 3.6

DATI: R = 1kΩ, V_{ON} = 0.5V

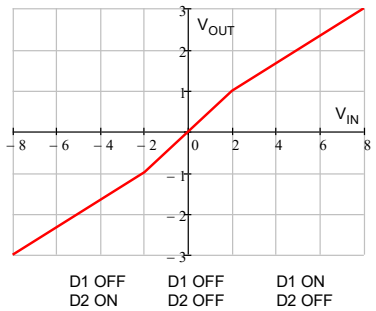
Dato il circuito in figura:

1. Tracciare la transcaratteristica.
2. Calcolare la tensione V_{OUT} con: V_{IN} = -5V, V_{IN} = 1V, V_{IN} = 5V



Soluzione

1. Transcaratteristica

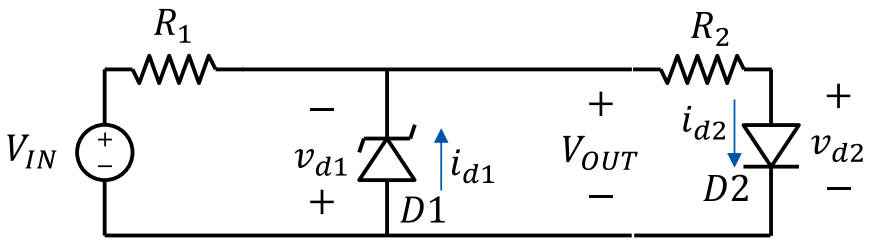
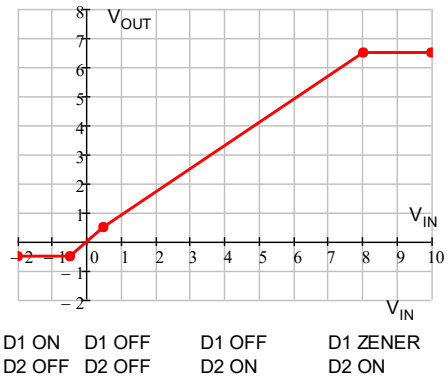


2. V_{OUT} = -2V, V_{OUT} = 0.5V, V_{OUT}=2V

Esercizio 3.7

DATI: R_S = 1kΩ, R_1 = 4kΩ, R_2 = 10kΩ, V_{ON} = 0, V_Z = 8V

Dato il circuito in figura, tracciare la transcaratteristica del circuito usando il teorema di Thevenin nel punto indicato dalla freccia.



Soluzione